

1

Governed Bilingual Inputs

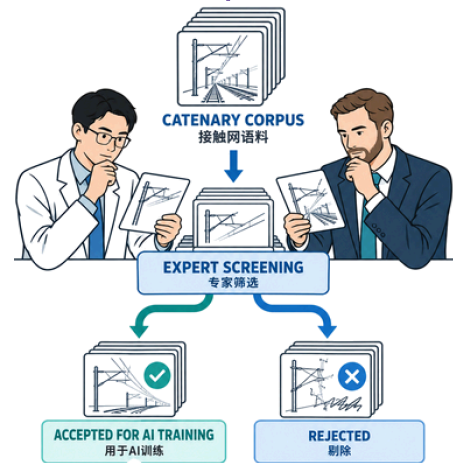


q_{zh} : 接触网供电的额定电压是多少

parallel inputs

q_{en} : What is the rated voltage for contact network power supply

Governed bilingual records
Chinese/English · source review · split



Admissibility filter
approved / test excluded

Query-time approval only; not a trainable mask.

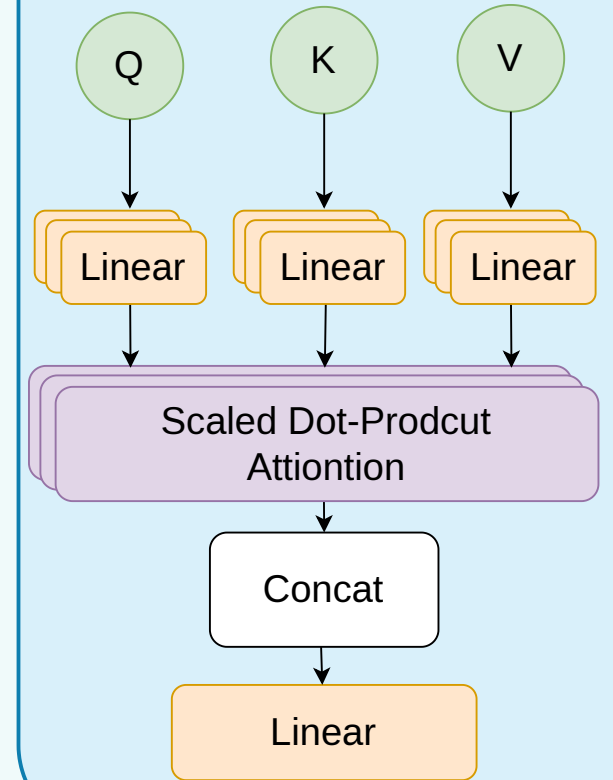
2

Shared Neural Encoding

Frozen shared BGE-M3 encoder
shared query/evidence parameters

Token + positional embeddings

Multi-Head Self-Attention



Residual + Add & Norm

Feed-Forward Network

Residual + Add & Norm x N

$$z_q = \text{Encoder}_{\text{BGE-M3}}(q)$$

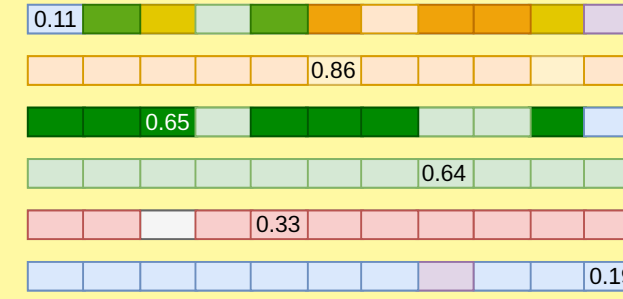
$$z_d = \text{Encoder}_{\text{BGE-M3}}(d)$$

$$z_q, z_d \in \mathbb{R}^{1024}$$

3

Parallel Hybrid Retrieval

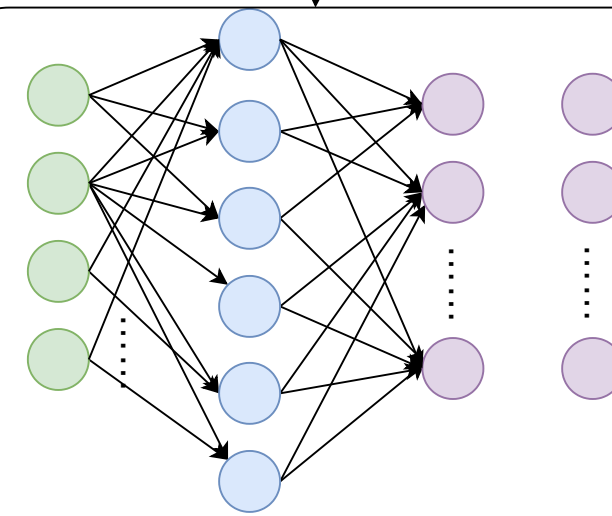
Embedding Matrix



Dense retrieval
 $\cos(z_q, z_d)$
vector rank

BM25 retrieval
lexical rank

Reciprocal Rank Fusion
 $\text{RRF}(d) = \sum 1/(k_0 + \text{rank})$



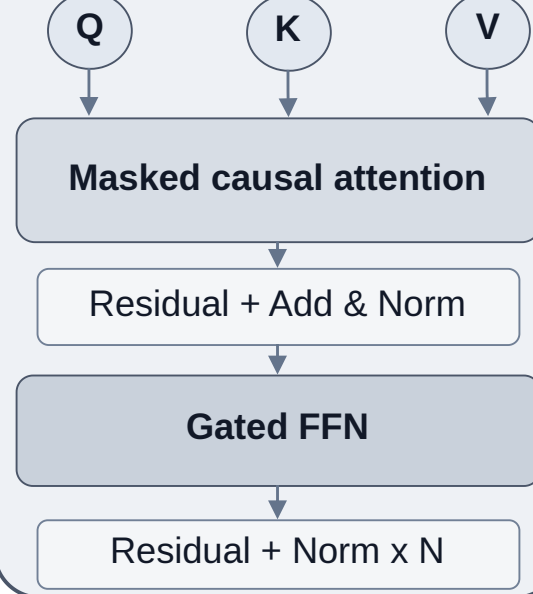
Top-k evidence passages
[E1]/[E2] passage + source

4

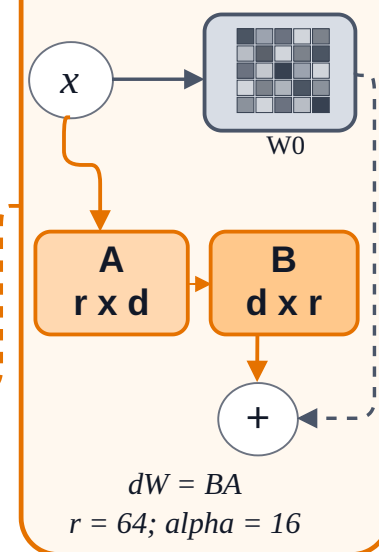
Evidence-Conditioned LLM + QLoRA

Serialized context
[Instr] | [Q] | evidence | [A]

Repeated decoder



QLoRA
low-rank update



One shared LM head + Softmax
 $p(y_t | X, y_{<t})$

Autoregressive bilingual answer
token1 -> token2 -> token3 -> ...
evidence labels: [E1] / [证据1]

Independent evaluation
external metrics / no gradient

Railway QA
F1 / containment

Citations
coverage / syntax

Translation
BLEU / chrF / COMET

Evidence support
semantic proxy

Resource-constrained inference
latency / GPU memory / tokens/s